

**5 In this question you must show detailed reasoning.**

The complex number  $w$  is given by  $w = -4\sqrt{2} + (4\sqrt{2})i$ .

**(a) (i)** Find  $|w|$ . **[2]**

**(ii)** Find  $\arg(w)$ . **[2]**

The complex numbers  $z_1$  and  $z_2$  are given by  $z_1 = a + i$  and  $z_2 = 4(\cos \theta + i \sin \theta)$ , where  $a$  is a positive real constant and  $-\pi < \theta \leq \pi$ .

**(b)** You are given that  $z_1 z_2 = w$ .

**(i)** Find the exact value of  $a$ . **[3]**

**(ii)** Find the value of  $\theta$ . Give your answer as an exact multiple of  $\pi$ . **[3]**